COMP 3613 Assignment 2

Individual report

Student information

Field	Details
Student name	Eliana Dookhoo
Student ID	816044896
Project	MyAdvisor
Repository	EllieD27/uwi-comp3613a1-my-advisor
Deployed application	https://faststarter-ejhk.onrender.com

1. Project description

MyAdvisor is an academic planning application for students and advisors. Students can view degree progress, build and submit semester plans, and review feedback. An administrator/advisor can view submitted plans, approve a plan, or request changes with a non-empty comment.

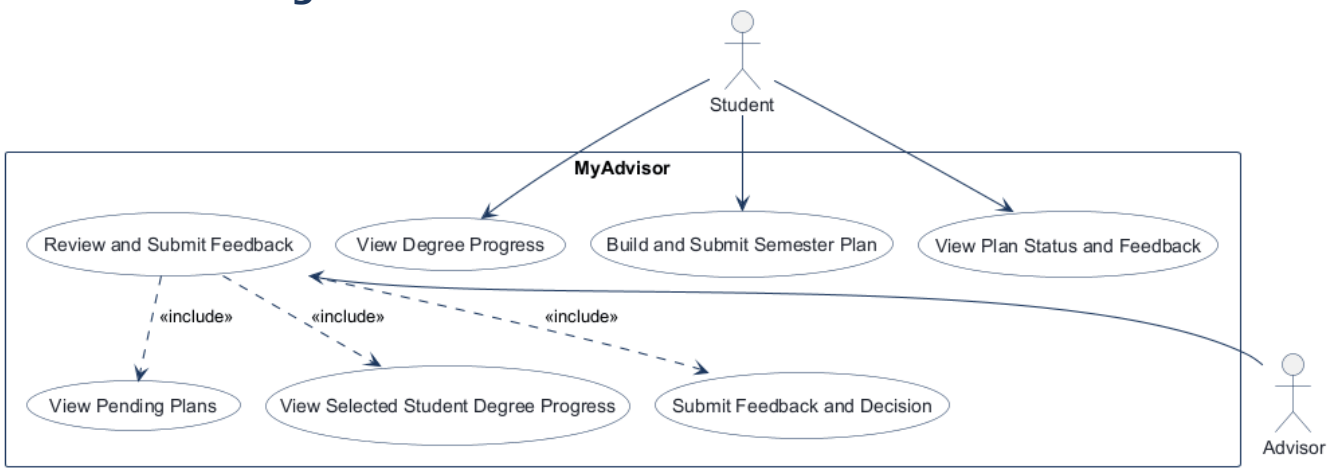
The minimum feature set is:

1. View Degree Progress (Student)
2. Build and Submit Semester Plan (Student)
3. Review and Submit Feedback (Advisor)

The deployed marker accounts are:

Username	Password	Role	Use
admin	adminpass	Administrator/advisor	Review and feedback workflow
bob	bobpass	Student	Student workflows
celeste	celestepass	Student	Student workflows
kate	katepass	Student	Student workflows
nelly	nellypass	Student	Student workflows
ian	ianpass	Student	Student workflows

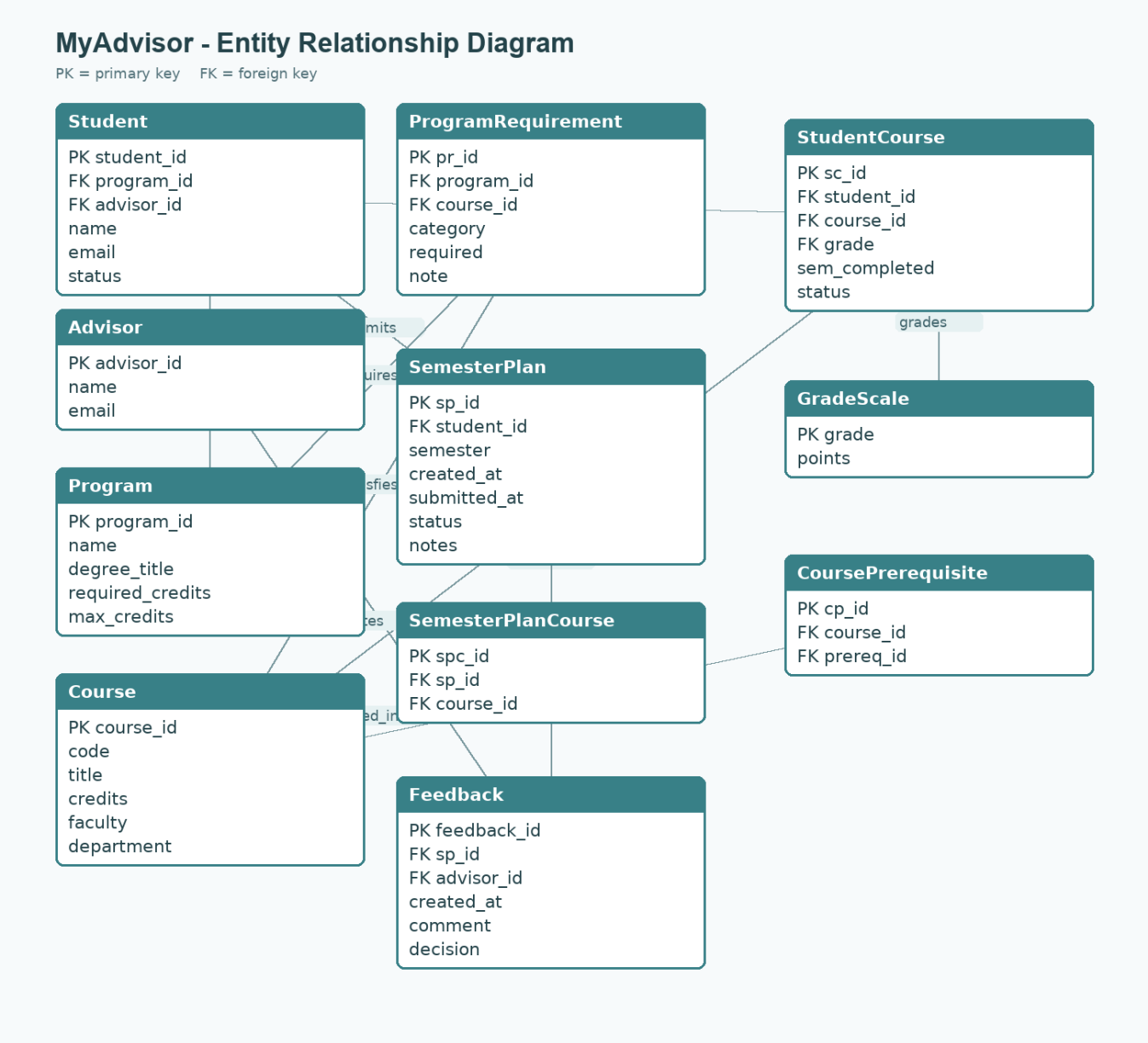
2. Use-case diagram



The editable source is use-case.json.

The advisor review workflow includes viewing pending plans, viewing the selected student's degree progress, and submitting feedback and a decision. Login is a precondition for protected student and advisor workflows.

3. Entity relationship diagram



The editable model source is model.mmd.

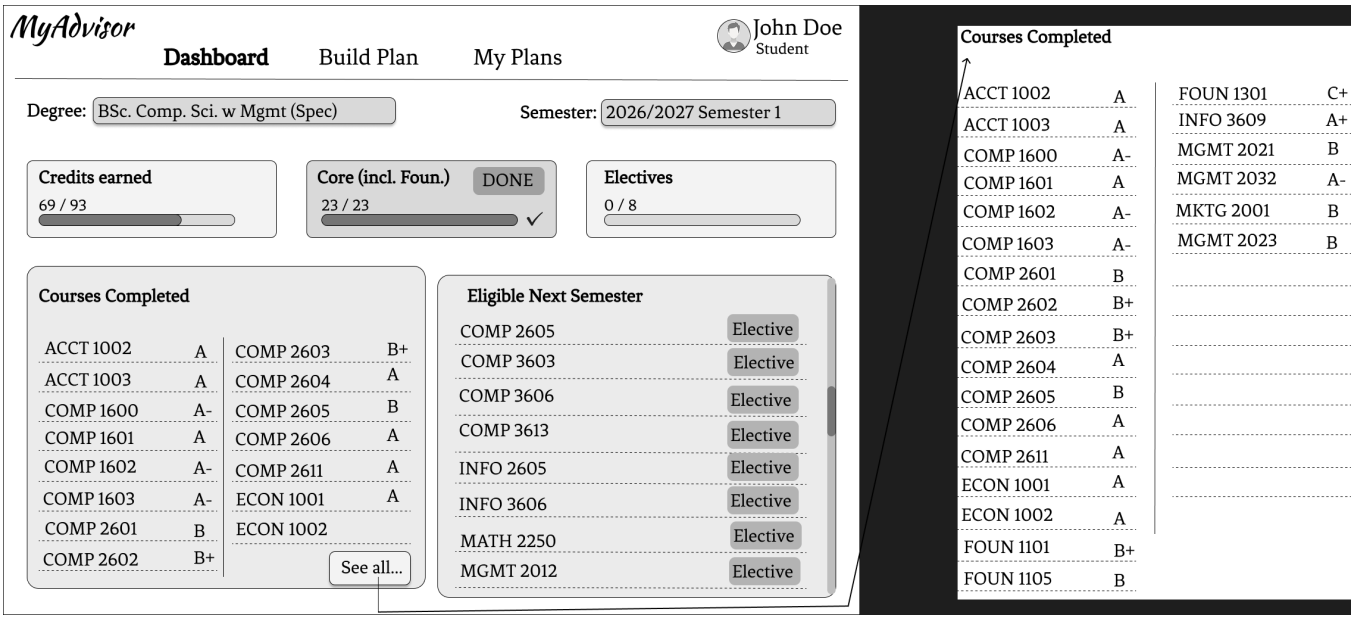
The implemented model includes:

- Student, Advisor, and Program
- Course, ProgramRequirement, and GradeScale
- SemesterPlan and SemesterPlanCourse
- StudentCourse and CoursePrerequisite
- Feedback

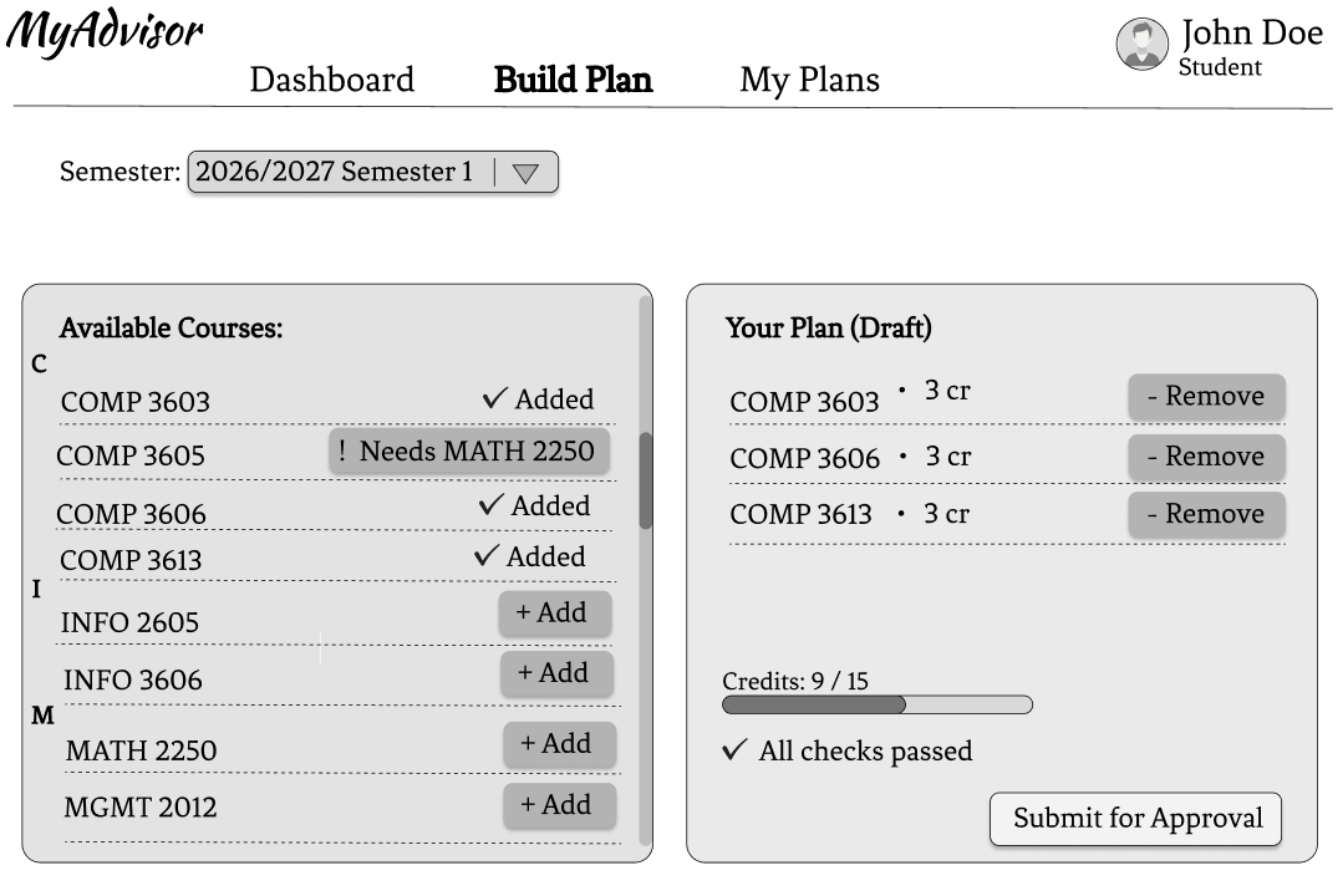
Important rules include a unique (sp_id, course_id) plan-course pair, credit-weighted GPA, prerequisite validation using completed C-or-better courses, plan credit limits, and advisor ownership checks.

4. Wireframes

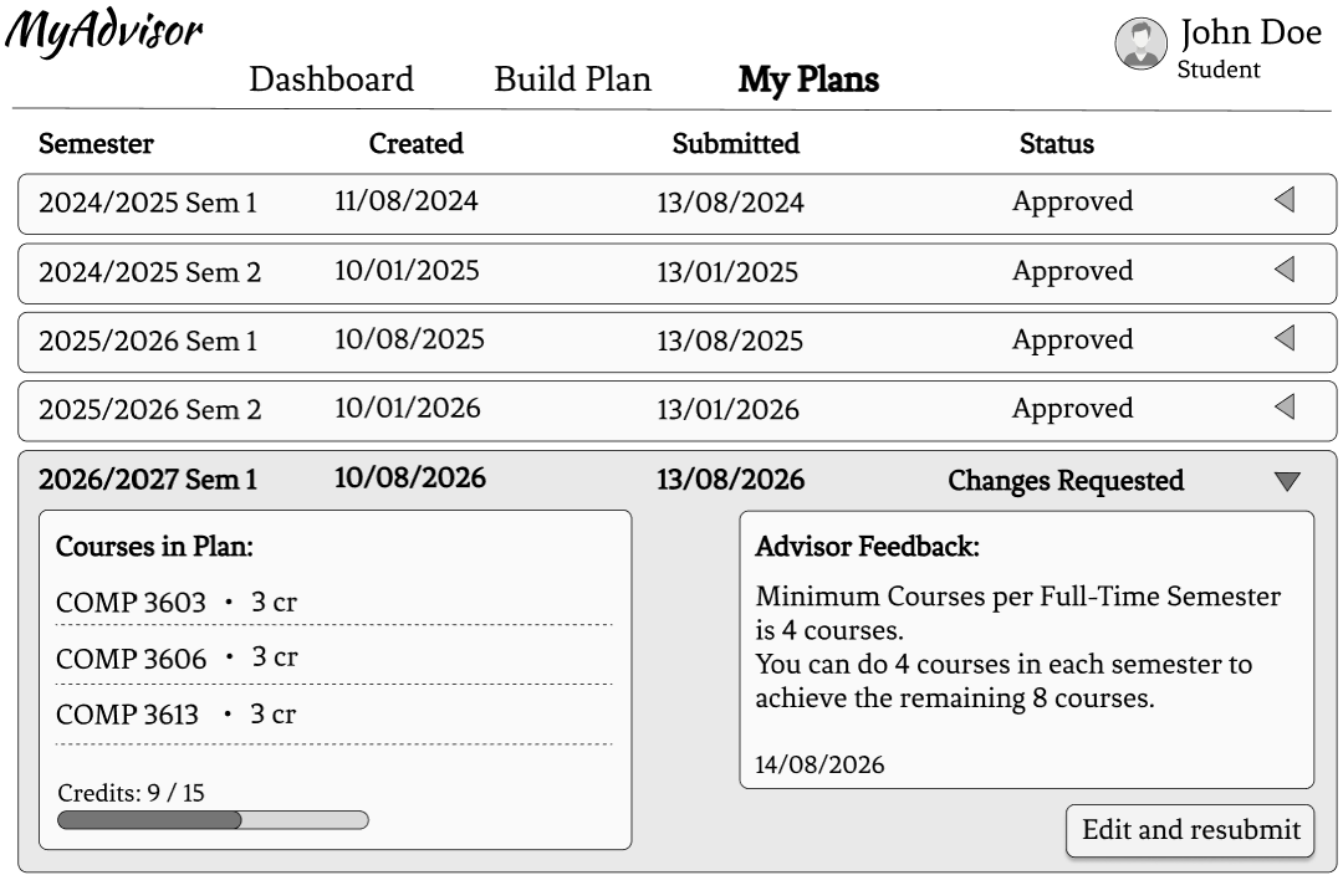
Student dashboard - View Degree Progress



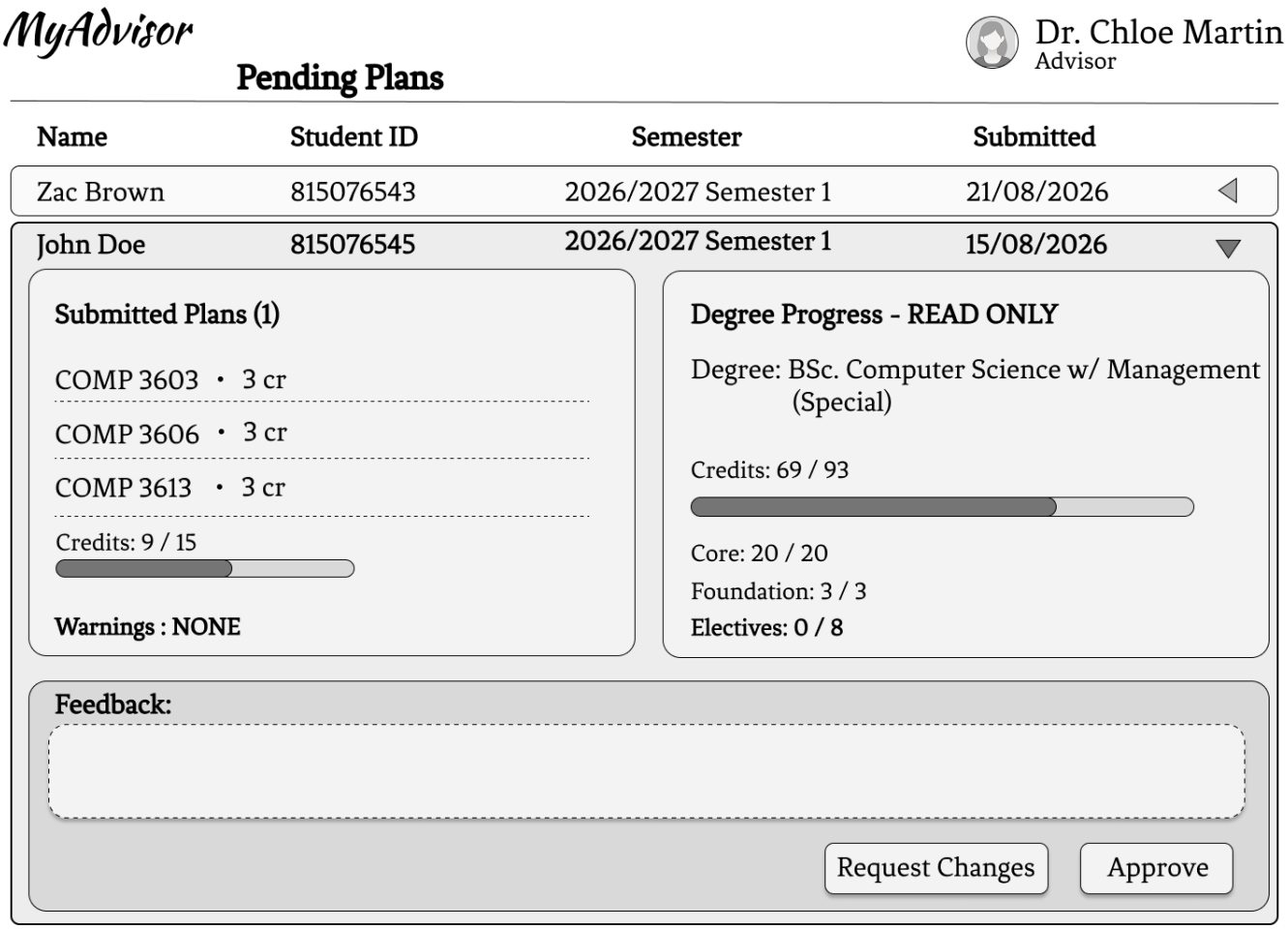
Student Build Plan - Build and Submit Semester Plan



Student My Plans - Review submitted plan and feedback



Advisor Review Plans - Review and decide on a semester plan



5. Test plans

The complete test-plan source is also retained in a2-test-plans.md.

5.1 Unit test plan

Unit Test	Input	Expected Output
test_academic_er_draft_models_exist_with_expected_keys	Academic model classes	All required ERD fields exist on the model classes
test_semester_plan_course_prevents_duplicate_course_entries	A SemesterPlanCourse table definition	A unique constraint exists for sp_id and course_id
test_feedback_decision_is_restricted_to_erd_values	The Feedback table definition	The decision constraint allows Approve and Request changes
test_passed_course_ids_requires_completed_c_or_better	Student course records with C, B, F, and in-progress grades	Only completed courses with C-or-better grades are returned

Unit Test	Input	Expected Output
test_dashboard_credit_and_course_count_cards_reconcile_to_93_credits	A student with completed core courses	Earned credits and core/elective counts match the degree requirements
test_gpa_is_credit_weighted_and_ignores_ungraded_courses	Completed A and B courses plus an ungraded in-progress course	GPA is calculated as a credit-weighted 3.5 and ignores the ungraded course
test_build_plan_service_marks_prerequisite_gap_and_selected_courses	Student plan, catalog courses, and prerequisite data	Selected courses, credits, course statuses, and prerequisite checks are correct
test_build_plan_catalog_is_bob_completed_and_planned_courses_in_code_order	The semester-plan catalog	The catalog contains the expected courses in code order
test_completed_course_is_marked_and_cannot_be_added_again	A completed course submitted for addition	The course is marked completed and is not added to the plan
test_semester_plan_service_add_and_submit_update_plan_state	A valid course addition followed by plan submission	The course is added and the plan status changes to Submitted
test_add_course_rejects_plan_that_would_exceed_credit_limit	A plan at the maximum credit limit	The additional course is rejected and no database object is added
test_add_course_rejects_course_when_prerequisite_is_not_passed	A course whose prerequisite is not passed	The course is rejected and no database object is added
test_my_plans_view_includes_erd_related_courses_and_feedback	A submitted plan with courses and advisor feedback	The view contains the plan, courses, credits, and feedback
test_my_plans_view_is_empty_for_user_without_student_record	A user with no student record	The service returns an empty plans view

Unit Test	Input	Expected Output
test_admin_dashboard_includes_assigned_pending_plan_and_gpa	An advisor with an assigned submitted plan	The dashboard includes the plan, student, courses, credits, progress, and GPA
test_submit_plan_decision_saves_feedback_and_updates_plan_status	A valid advisor decision and comment	Feedback is saved and the plan status is updated
test_submit_plan_decision_rejects_unassigned_or_invalid_decisions	An invalid decision or an unassigned advisor	No feedback or plan update is saved
test_request_changes_requires_a_comment	Request changes with an empty comment	The decision is rejected and the plan remains submitted
test_advisor_review_template_contains_wireframe_details_and_gpa	Advisor review template source	Required review controls, plan details, and GPA labels are present
test_build_plan_course_options_are_scrollable	Build-plan template source	Course options have the required scrollable layout

The consolidated implementation is tests/test_unit.py.

5.2 Integration test plan

Test	Dependencies	Description
test_health	Test client, health route	Calls GET /health and verifies the application health endpoint returns HTTP 200
test_public_landing_page_is_available	Test client, in-memory database, landing route	With the isolated test database and no login, verifies that the public landing page returns HTTP 200 and displays MyAdvisor
test_student_dashboard_requires_login	Test client, login route, authentication dependency	With no authentication cookie, verifies that the student dashboard redirects to the login page

Test	Dependencies	Description
test_authenticated_student_can_view_dashboard	Test client, login route, seeded student, in-memory database	Logs in as the seeded student and verifies that the student dashboard returns HTTP 200 with degree-progress content
test_regular_user_cannot_view_admin_dashboard	Test client, login route, seeded student, admin dependency	Logs in as a regular student and verifies that access to the admin dashboard returns HTTP 401
test_admin_can_view_admin_dashboard	Test client, login route, seeded admin, in-memory database	Logs in as the seeded admin and verifies that the admin dashboard returns HTTP 200
test_users_api_returns_user_response_shape	Test client, login route, admin dependency, user repository	Verifies unauthenticated GET /api/users is rejected, then verifies an authenticated admin receives only the id/username/email response fields
test_student_can_add_and_submit_a_semester_plan	Test client, login route, plan service, in-memory database	Logs in as a student, adds COMP 3603, submits the plan, and verifies the database status is Submitted
test_admin_can_decide_on_submitted_semester_plan	Test client, login route, plan service, advisor review service, in-memory database	Creates a submitted student plan, logs in as admin, approves it, and verifies the plan status and feedback record
test_student_cannot_post_an_advisor_decision	Test client, student login, admin dependency, advisor review service	A student attempts to approve a submitted plan and the test verifies authorization failure, unchanged plan status, and no feedback

Test	Dependencies	Description
test_request_changes_without_a_comment_is_rejected	Test client, admin login, advisor review service, feedback validation	An admin submits Request changes with an empty comment and the test verifies the plan remains Submitted with no feedback
test_course_with_unmet_prerequisite_cannot_be_added	Test client, student login, plan service, prerequisite data	A student attempts to add COMP 3605 without its prerequisite and the test verifies no plan-course row is created
test_advisor_cannot_decide_on_another_advisors_student	Test client, admin login, advisor assignment data, advisor review service	An admin attempts to decide on a plan assigned to another advisor and the test verifies the plan remains Submitted with no feedback

The repository-focused unit test is an exception to the usual isolated unit boundary: it uses an in-memory SQLite database. The two template-source unit tests are also exceptions because they read template files. The remaining unit tests use mocked or in-memory service inputs without the application test client.

All integration tests use the isolated in-memory SQLite fixture in tests/conftest.py; they do not modify the development database. The implementation is in tests/test_integration.py.

5.3 User acceptance test report

Tester Name: Eliana Dookhoo **Tester Position:** Software Engineering 2 Student **Date of Test:** 10 / 11 / 2026

Test Case	Pre-conditions	Test Steps	Test Criteria	Success (indicate with a check)
View Degree Progress	The student account exists and the tester is logged in as a student.	1. Open the student dashboard; 2. Review the degree-progress cards.	The dashboard shows earned credits, required credits, core/elective course counts, and GPA.	☑ Pass / ☐ Fail

Test Case	Pre-conditions	Test Steps	Test Criteria	Success (indicate with a check)
Build and Submit Semester Plan	The tester is logged in as a student and the course catalogue is available.	1. Open Build Plan; 2. Select a valid course; 3. Add the course to the plan; 4. Submit the plan.	The selected course and total credits are displayed, the submission completes, and the plan status becomes Submitted.	☑ Pass / ☐ Fail
Review and Approve Semester Plan	A student plan is submitted and the tester is logged in as the assigned advisor/admin.	1. Open the advisor dashboard; 2. Review the submitted plan; 3. Enter feedback; 4. Select Approve.	The plan is removed from the pending queue, the plan status becomes Approved, and the feedback is saved.	☑ Pass / ☐ Fail
Request Changes to a Semester Plan	A student plan is submitted and the tester is logged in as the assigned advisor/admin.	1. Open the submitted plan; 2. Enter a non-empty feedback comment; 3. Select Request Changes.	The plan status becomes Request Changes and the feedback is available for the student to review.	☑ Pass / ☐ Fail
Prevent Invalid Plan Actions	The tester is logged in as a student and selects a course whose prerequisite has not been passed.	1. Open Build Plan; 2. Attempt to add the course with the unmet prerequisite.	The course is not added, the plan remains unchanged, and the course status explains that a prerequisite is required.	☑ Pass / ☐ Fail

Acceptance Decision: ☑ Accepted / ☐ Rejected

Signature: Eliana Dookhoo

6. Implementation

6.1 Repository and architecture

Repository: [EllieD27/uwi-comp3613a1-my-advisor]

The application uses FastAPI and SQLModel with the layered architecture:

Routes -> Dependencies -> Schemas -> Services -> Repositories -> Models

Routes bind request data and call services. Repositories contain persistence operations, while services contain application rules such as prerequisite validation, credit limits, plan status changes, and advisor ownership.

6.2 Feature implementation

Req Num	Feature	Controller / Route	Description
0	Authentication and authorization	app/routers/login.py, AuthDep, AdminDep	Authenticates users, preserves safe redirects, routes students to /app, routes admins to /admin, and protects role-specific pages.
1	View degree progress	app/routers/user_home.py, DegreeProgressService	Displays earned credits, required credits, course counts, prerequisites, and credit-weighted GPA for the signed-in student.
2	Build and submit semester plan	app/routers/user_home.py, SemesterPlanService	Creates or loads a student plan, validates course selection and prerequisites, enforces credit limits, and submits the plan.

Req Num	Feature	Controller / Route	Description
3	Review and approve semester plan	app/routers/admin_home.py, AdvisorReviewService	Allows the assigned advisor/admin to review a submitted plan and approve it with feedback.
4	Request changes	app/routers/admin_home.py, AdvisorReviewService	Requires a non-empty comment, records feedback, and changes the plan status to Request changes.
5	Prevent invalid plan actions	SemesterPlanService, AdvisorReviewService	Rejects duplicate courses, completed courses, unmet prerequisites, excessive credits, unauthorized decisions, and blank change requests.

6.3 Authorization

- Student routes use the authenticated-user dependency.
- Admin routes use the admin dependency.
- Unauthenticated student dashboard access redirects to /login.
- Unauthenticated admin access returns HTTP 401.
- Non-admin users cannot submit advisor decisions.
- Advisors cannot decide on plans belonging to another advisor's student.

6.4 REST/API implementation

The collection tests these endpoints:

Method	Endpoint	Purpose
GET	/health	Health check
GET	/	Public landing page
GET	/api/users	Admin-only user response-shape test

Method	Endpoint	Purpose
GET	/app	Protected student dashboard
GET	/app/build-plan	Authenticated student course-planning page
POST	/app/build-plan	Authenticated student add/remove/submit plan actions
GET	/app/my-plans	Authenticated student submitted-plan history
GET	/admin	Protected administrator dashboard
POST	/admin/plans/{id}/decision	Authenticated advisor approval or change request
POST	/login	Role-based login routing

Limitations. The workflow endpoints use the existing server-rendered form routes rather than a separate JSON API. They still follow resource-oriented paths and return redirects after successful form mutations. The Postman collection authenticates a student and an administrator, exercises the student plan workflow, and submits an administrator decision. A future API revision could add JSON representations without changing these workflow services.

6.5 API specification

Description	Method	URL	Request / Headers	Response / Status
Health check	GET	/health	None	{"ok": true} / 200
Public landing page	GET	/	None	Landing page HTML / 200

Description	Method	URL	Request / Headers	Response / Status
Admin users response	GET	/api/users	Authenticated admin session cookie	User fields as an array / 200; unauthenticated or non-admin access is rejected
Student dashboard	GET	/app	Authenticated session cookie	Dashboard HTML / 200; unauthenticated access redirects to /login / 303
Build-plan page	GET	Authenticated student session cookie	Student build-plan HTML / 200	
Add course	POST	/app/build-plan	Authenticated student cookie; form action=add&course_code=COMP 3603	Redirect to /app/build-plan / 303; valid course is added
Submit plan	POST	/app/build-plan	Authenticated student cookie; form action=submit	Redirect to /app/build-plan / 303; plan status becomes Submitted
My plans	GET	Authenticated student session cookie	Student plan history HTML / 200	
Admin dashboard	GET	/admin	Authenticated admin session cookie	Dashboard HTML / 200; unauthenticated or non-admin access / 401

Description	Method	URL	Request / Headers	Response / Status
Plan decision	POST	/admin/plans/{id}/decision	Authenticated admin cookie; form decision=Approve and comment=...	Redirect to /admin / 303; plan status becomes Approved
Login	POST	/login	Form username and password	Role-based redirect to /app or /admin / 303; invalid credentials return the login page with an error

7. Unit-test implementation

Existing unit tests were retained and expanded for Assignment 2. The consolidated unit-test module is:

• test_unit.py

Run the unit tests with:

```
.\.venv\Scripts\python.exe -m pytest tests\test_unit.py -v --color=yes
```

8. Integration-test implementation

Run the integration tests with:

```
.\.venv\Scripts\python.exe -m pytest tests\test_integration.py -v --color=yes
```

The fixture in tests/conftest.py creates an isolated database, seeds separate student and admin identities, and overrides the application session dependency so the development database is not modified.

9. Postman collection

The exported Postman v2.1 collection is:

MyAdvisor-Assignment-2-API-Tests.postman_collection.json

Collection name:

```
MyAdvisor Assignment 2 API Tests
```

The collection contains:

• Health and public endpoint tests
• Admin-only users API response-shape and refusal tests
• Unauthenticated student and admin authorization tests

- Admin login routing test
- Student login routing test
- Authenticated student plan-page, add-course, submit-plan, and my-plans requests
- Authenticated admin plan-decision request

The baseUrl collection variable defaults to:

http://127.0.0.1:5000

For the deployed application, change it to:

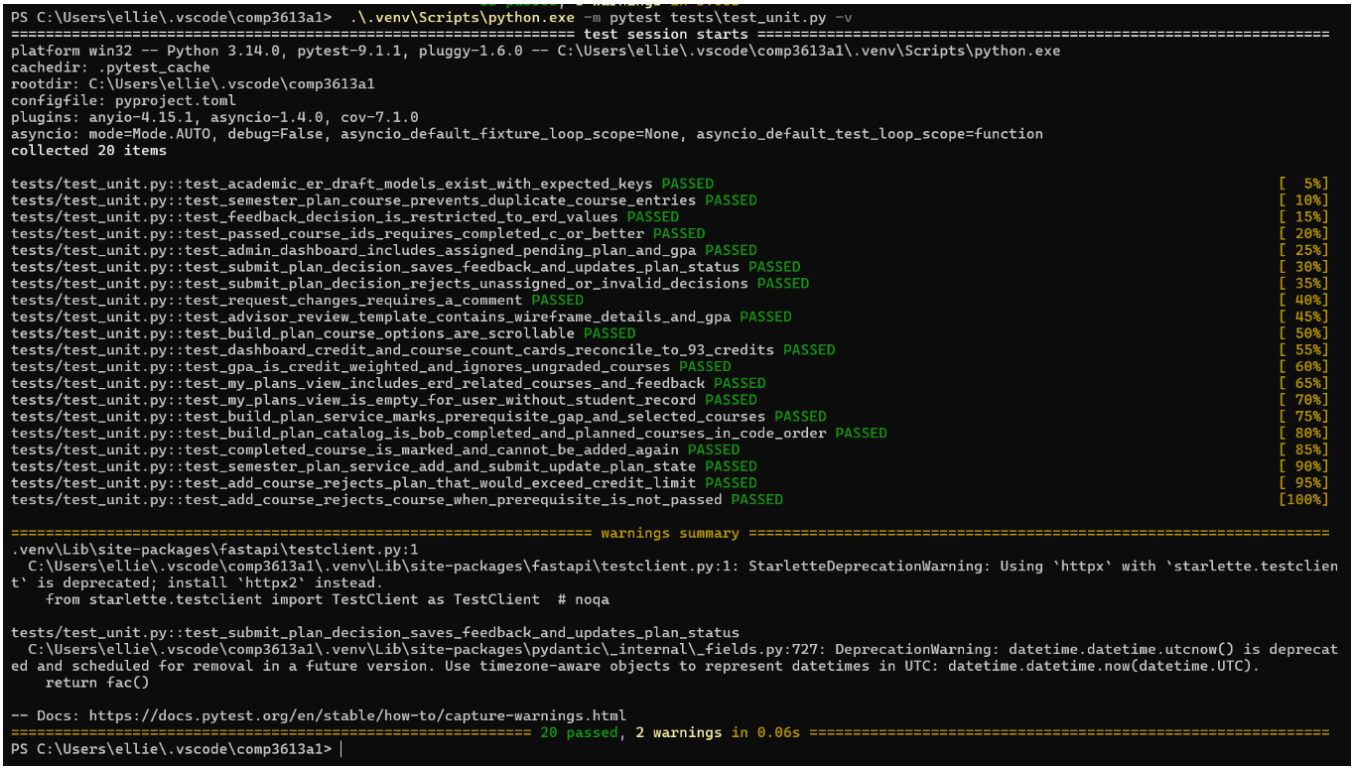
https://faststarter-ejhk.onrender.com

The authorization requests disable redirects and isolate the cookie state so consecutive collection runs do not reuse a successful login session.

10. Test-report evidence

The Postman collection was run locally with one iteration. The run completed with 23 passing tests, 0 failures, 0 skipped tests, and 0 errors.

Unit-test screenshot



Integration-test screenshot

```
PS C:\Users\ellie\.vscode\comp3613a1> .\.venv\Scripts\python.exe -m pytest tests\test_integration.py -v --color=yes
============================================================= test session starts ==============================================================
platform win32 -- Python 3.14.0, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\ellie\.vscode\comp3613a1\.venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\ellie\.vscode\comp3613a1
configfile: pyproject.toml
plugins: anyio-4.15.1, asyncio-1.4.0, cov-7.1.0
asyncio: mode=Mode.AUTO, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 13 items

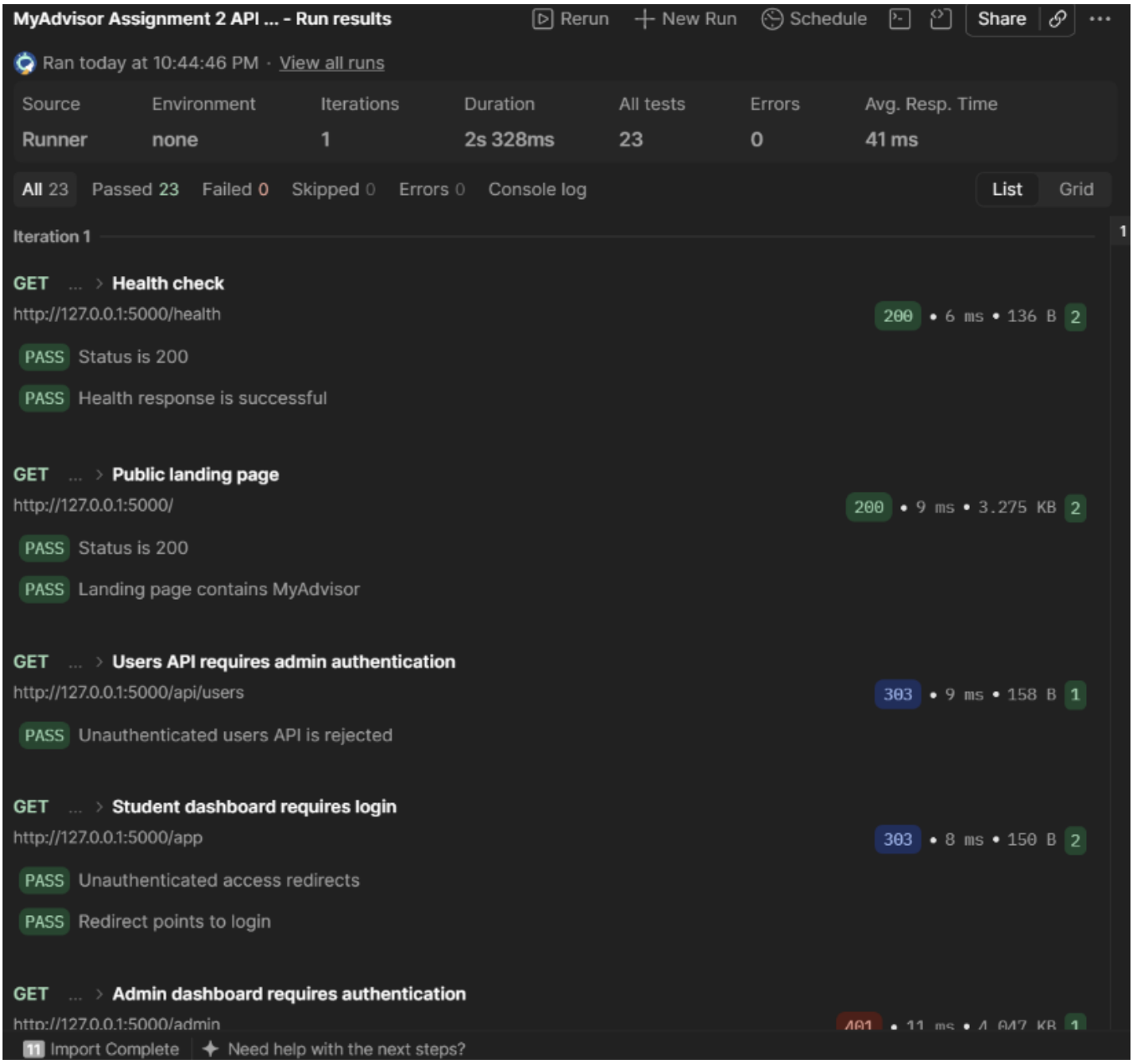
tests/test_integration.py::test_health PASSED                                                                                          [  7%]
tests/test_integration.py::test_public_landing_page_is_available PASSED                                                                [ 15%]
tests/test_integration.py::test_student_dashboard_requires_login PASSED                                                                [ 23%]
tests/test_integration.py::test_authenticated_student_can_view_dashboard PASSED                                                        [ 30%]
tests/test_integration.py::test_regular_user_cannot_view_admin_dashboard PASSED                                                        [ 38%]
tests/test_integration.py::test_admin_can_view_admin_dashboard PASSED                                                                  [ 46%]
tests/test_integration.py::test_users_api_returns_user_response_shape PASSED                                                           [ 53%]
tests/test_integration.py::test_student_can_add_and_submit_a_semester_plan PASSED                                                      [ 61%]
tests/test_integration.py::test_admin_can_decide_on_submitted_semester_plan PASSED                                                     [ 69%]
tests/test_integration.py::test_student_cannot_post_an_advisor_decision PASSED                                                         [ 76%]
tests/test_integration.py::test_request_changes_without_a_comment_is_rejected PASSED                                                   [ 84%]
tests/test_integration.py::test_course_with_unmet_prerequisite_cannot_be_added PASSED                                                  [ 92%]
tests/test_integration.py::test_advisor_cannot_decide_on_another_advisors_student PASSED                                               [100%]

============================================================== warnings summary ================================================================
.venv\Lib\site-packages\fastapi\testclient.py:1
  C:\Users\ellie\.vscode\comp3613a1\.venv\Lib\site-packages\fastapi\testclient.py:1: StarletteDeprecationWarning: Using `httpx` with `starlette.testclien
t` is deprecated; install `httpx2` instead.
    from starlette.testclient import TestClient as TestClient  # noqa

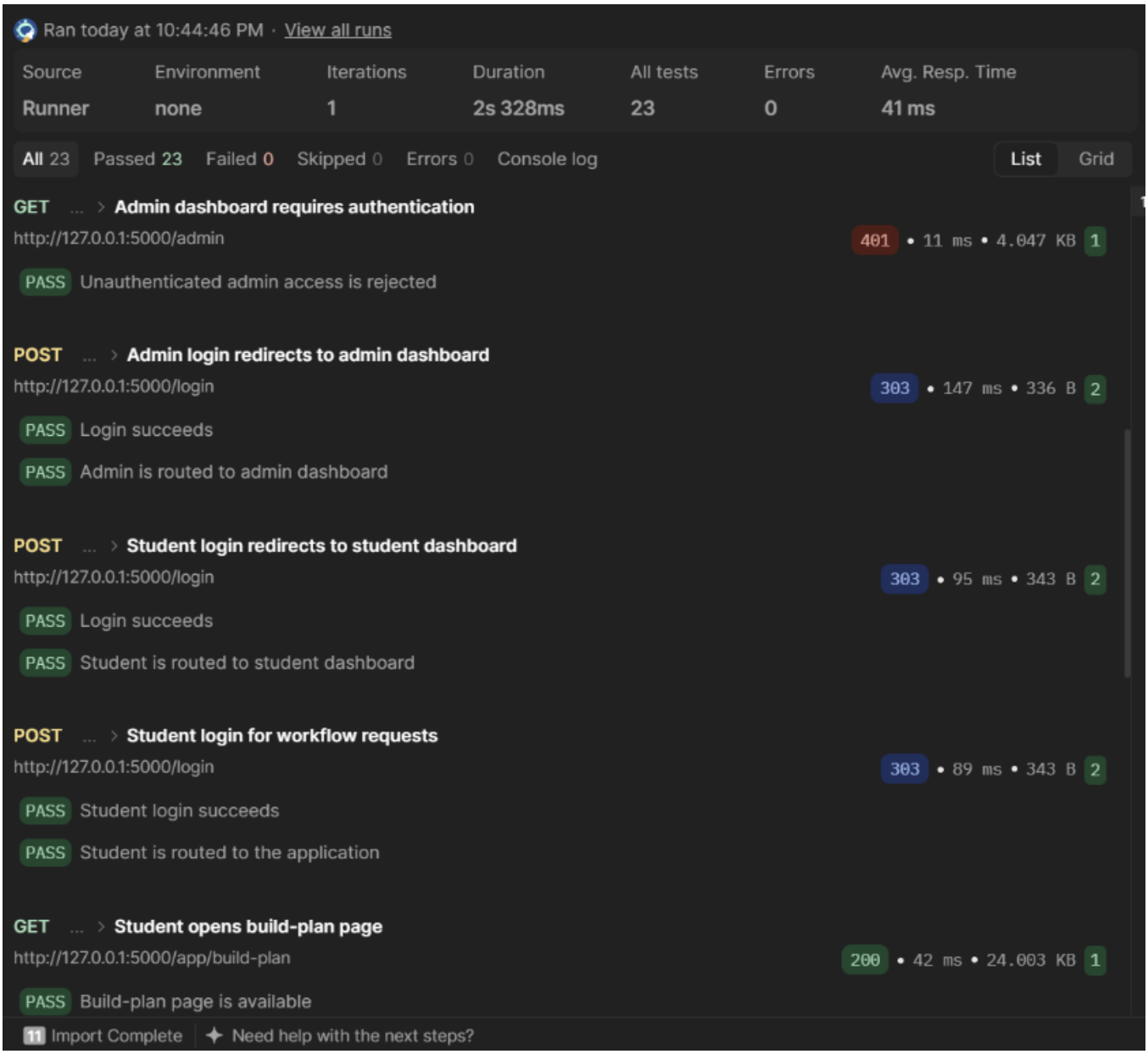
tests/test_integration.py::test_student_can_add_and_submit_a_semester_plan
tests/test_integration.py::test_admin_can_decide_on_submitted_semester_plan
tests/test_integration.py::test_admin_can_decide_on_submitted_semester_plan
tests/test_integration.py::test_student_cannot_post_an_advisor_decision
tests/test_integration.py::test_request_changes_without_a_comment_is_rejected
tests/test_integration.py::test_course_with_unmet_prerequisite_cannot_be_added
tests/test_integration.py::test_advisor_cannot_decide_on_another_advisors_student
  C:\Users\ellie\.vscode\comp3613a1\.venv\Lib\site-packages\pydantic\_internal\_fields.py:727: DeprecationWarning: datetime.datetime.utcnow() is deprecat
ed and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UC: datetime.datetime.now(datetime.UTC).
    return fac()

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
============================================================ 13 passed, 8 warnings in 4.25s ============================================================
PS C:\Users\ellie\.vscode\comp3613a1>
```

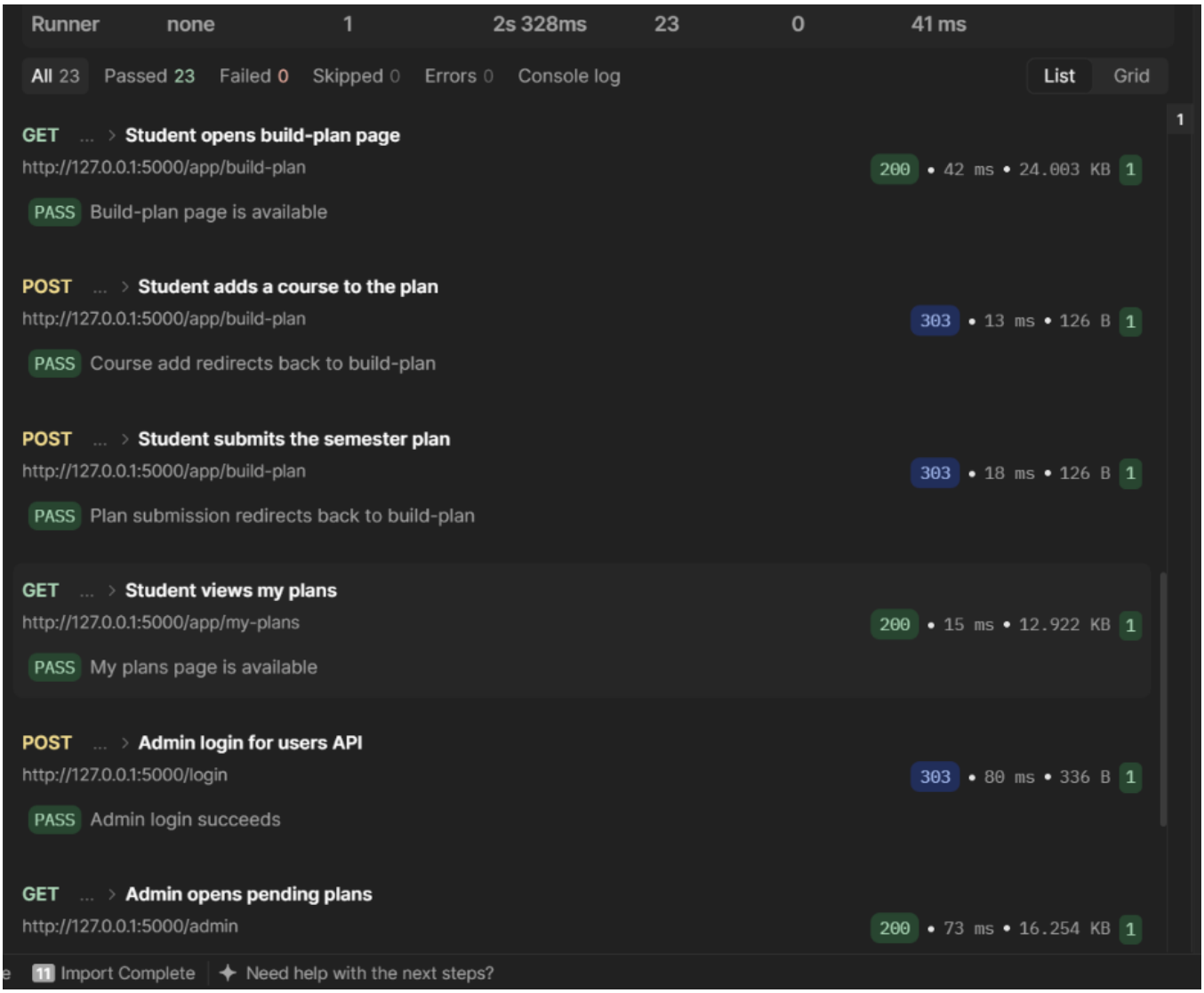
Postman collection screenshot



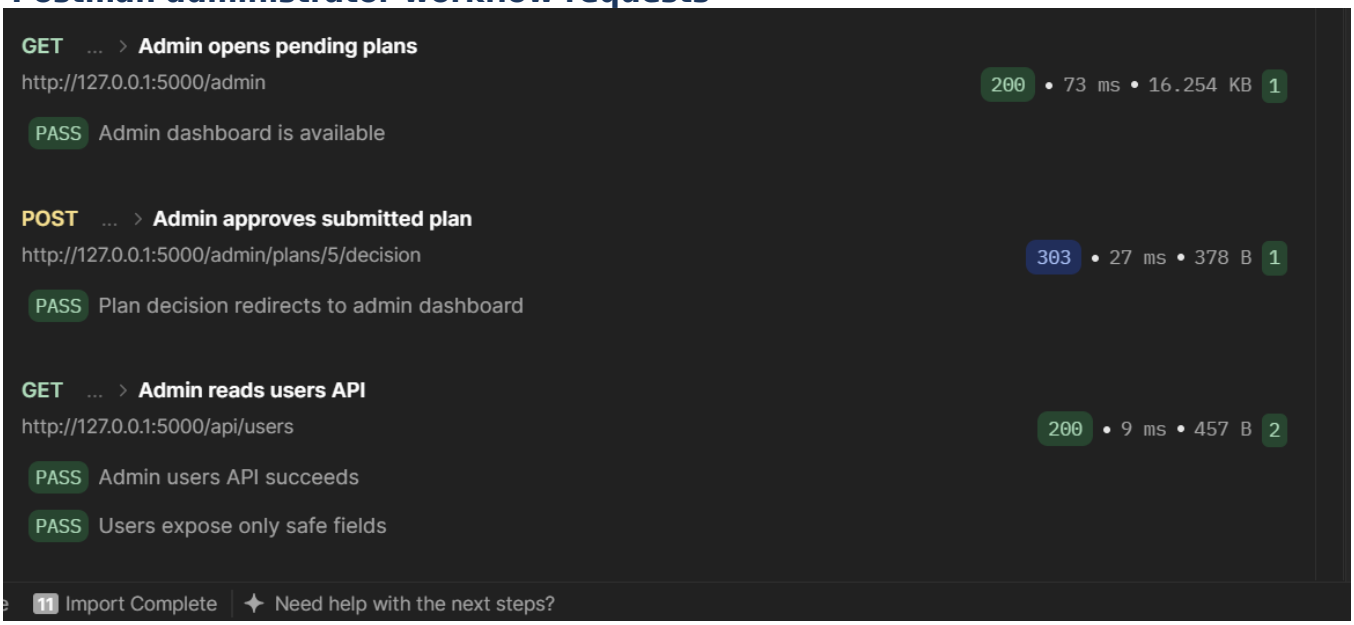
Postman authorization and workflow requests



Postman student workflow requests



Postman administrator workflow requests



11. UAT evidence

The completed UAT form is included in a2-test-plans.md, including the five passing cases,

tester details, accepted decision, and signature.